

PAULO STROBEL SILVA, PhD

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WORK EXPERIENCE

Research Fellow - Multiphysics Software Development (Cranfield University - AIRBUS) Oct 2022 - Present

- Developed comprehensive multiphase CFD software ecosystem for landing gear shock absorber simulation with model-to-experiment accuracy of < 12% damping force prediction
- Automated framework of multiple open-source software on HPC infrastructure, enabling parametric studies on mesh refinement and multiphase modeling
- Engineered Python workflows enabling coupled multiphysics simulation across multiple fidelity levels (low-order, steady-state, multiphase) with 3 orders of magnitude computational time reduction
- Supervised 10 MSc engineers on multidisciplinary research projects, establishing best practices for technical mentoring and thesis project scoping
- Delivered comprehensive training program for AIRBUS personnel on software utilization and underlying physics

CFD Engineer Consultant (Independent CFD Consultant) Jan 2012 - Present

- **Automated workflow for Commercial and open-source softwares**
- **Data Center Thermal Optimization:** Delivered +240–270% RCI improvements for 10+ hyperscale facilities (Scala: Santa Tereza, Chile(2.5MW); Curauma-Chile(5MW)). Optimized designs achieved 100% ASHRAE thermal compliance.
- **Critical Failure Analysis:** Conducted transient CFD simulations (30-second duration) for hyperscale facilities (1.6–5 MW), modeling equipment shutdown cascades to quantify thermal safety margins and inform redundancy requirements
- **Delivered high-impact technical reports** with safety compliance for multinational enterprises (Equinix, SCALA, Banco do Brasil)

Research Assistant (Cranfield University) Nov 2019 - Jan 2020; Nov 2021 - Sept 2022

- Delivered technical instruction for MSc CFD programme, mentoring 40+ graduate engineers in computational methodology and best practices
- Provided industrial CFD consultancy for jet optimization in microplastics cleaning applications, achieving 18% efficiency improvement

Research Assistant - University of Brasilia Mar 2012 - Jan 2016

- Pioneered multiphase CFD methodology for tidal turbine hydrodynamic optimization, achieving 12% power coefficient improvement through simulation-guided blade design
- Project recognized as “Project of the Year” by HydroVision Brasil and funded by Eletronorte (Brazilian Energy agency)
- Delivered technical simulation reports to project stakeholders and funding agencies

Mechanical Engineer, FOX Engenharia, Brasilia October 2013 - February 2014

- Thermal load calculations, HVAC system design and optimization for data center infrastructure
- Performed CFD simulations of operational data center HVAC systems, identifying hot spots and optimizing cooling strategies for energy cost reduction

EDUCATION

PhD in Aerospace Engineering - Cranfield University Oct 2017 - Jul 2022

- Pioneered rotating reference frame implementation in open-source Fortran CFD code for high-fidelity rotary wing simulations
- Validated computational models against experimental data: helicopter rotor wind tunnel tests (< 2% thrust coefficient error); NREL PHASE VI wind turbine benchmark (3–6% power prediction error); dynamic mesh motion for shock absorber simulations (< 8% damping force error)
- Developed Python/bash automation frameworks for geometry parametrization and mesh generation, accelerating design iterations by 70% reduction in setup time
- Published 5 peer-reviewed papers in high-impact journals (Aerospace Science & Technology, Energy)

University of Brasilia, Brazil (5th best university in Brazil, Times Higher Education, 2023) Brasilia, Brazil

MSc in Mechanical Sciences Mar 2012 - Mar 2014

- Thesis: “Numerical Study of Horizontal Axis Hydrokinetic Turbines” - focus on wake interactions and cavitation.

BSc in Mechanical Engineering Jul 2007 - Dec 2012

- Final project: CFD simulation of thermal comfort in indoor environments

TECHNICAL SKILLS

CFD Software ANSYS Fluent, ANSYS CFX, OpenFOAM · **Meshing** ANSYS ICEM CFD, Pointwise, SnappyHexMesh, Gmsh

Programming Python, FORTRAN, Bash, MATLAB · **CAD & Visualization** SolidWorks, AutoCAD, Tecplot, ParaView

Methodologies RANS (k- ϵ , k- ω SST), LES, Conjugate Heat Transfer, Multiphase Flow, Dynamic/Moving Meshes

Validation Experience Wind tunnel testing (rotors, turbines), Drop testing (shock absorbers), on-site facility measurements (data centers). Model to experiment accuracy: < 2% (helicopter rotors), 3–6% (wind turbines), < 12% (shock absorbers)

HPC & Automation High-Performance Computing, Cloud Computing, Workflow Automation, Parallel Processing

Languages Native Portuguese, fluent English (IELTS 7.5)

SELECTED PUBLICATIONS

Track record of 10+ peer-reviewed papers in top-tier journals. Full list: [Google Scholar](#)

• **Silva, P.A.S.F.**, et al. (2024). Study of orifice design on oleo-pneumatic shock absorber. *Fluids*, 9(5), 108.

• **Silva, P.A.S.F.**, et al. (2024). Tip vortex trajectory in shrouded wind turbines. *Energy*, 130929.

• **Silva, P.A.S.F.**, et al. (2022). Full helicopter with and without ground effect. *Aerospace Science and Technology*.